

PARTHA PRATIM BHAKAT

Machine Learning Engineer | Software Developer

Bhopal, India

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Professional Summary

Machine Learning Engineer and Computer Science undergraduate with hands-on experience in **deep learning, generative AI, and agentic LLM systems**, gained through a paid AI/ML internship, applied research, and an award-winning hackathon project. Seeking a full-time **Machine Learning Engineer / AI Engineer** role to apply model development and deployment skills to production-scale problems.

Technical Skills

Languages: Python, Java, SQL, HTML5, CSS3

Machine Learning & AI: Neural Networks (CNN, ANN), Deep Learning, Supervised & Unsupervised Learning, NLP, Computer Vision, Scikit-learn, TensorFlow, Keras, PyTorch, Hugging Face, OpenCV, LLMs, Generative AI, Agentic AI, OpenAI GPT-4o-mini, Prompt Engineering, Model Deployment

Data Analysis & Visualization: NumPy, Pandas, Matplotlib, Seaborn, Recharts

Web & Backend Development: Next.js, FastAPI, REST APIs, Pydantic

Database Management: MySQL, MongoDB, DBMS

Tools & Platforms: Git, GitHub, Docker, VS Code

Experience

AI/ML Intern

Mar 2026 – Apr 2026

Rigzz Technologies

Kolkata, West Bengal

- **Assisted** in building and testing machine learning models, contributing to data preprocessing and feature engineering that improved input data quality for downstream model evaluation.
- **Collaborated** with cross-functional engineers to debug and optimize AI/ML pipelines, reducing recurring pipeline errors through documented fixes and code reviews.
- **Participated** in daily stand-ups and sprint planning under agile practices, consistently delivering assigned components on schedule.

Projects

AI Text-to-SQL | Python, Google Gemini LLM, MySQL, Agentic AI

May 2026

- **Created** a Text-to-SQL chatbot using Google Gemini to translate natural language queries into executable MySQL statements, backed by an automated pipeline that ingests and maps raw Excel data into a structured database.
- **Constructed** an LLM chain with a custom output parser to dynamically execute queries and format results into conversational responses.
- **Implemented** an Agentic workflow and evaluated query generation accuracy and response fidelity using the RAGAS framework.

Real-Time ASL Detection 🗣️ | Python, OpenCV, TensorFlow, Keras

May 2026

- **Developed** a real-time American Sign Language detection application classifying 24 alphabet gestures via a live webcam feed.
- **Designed** and trained a custom CNN from scratch on the Sign Language MNIST dataset, achieving 99.8% training accuracy using batch normalization, dropout, and dynamic learning-rate reduction to prevent overfitting.
- **Integrated** OpenCV for real-time video processing, leveraging background subtraction and contour detection to isolate hand gestures.

Education

Vellore Institute of Technology-Bhopal

Aug 2023 – May 2027

B.Tech, Computer Science and Engineering – CGPA: 7.74

Bhopal, India

Publications & Patents

- **Patent:** Self-Recharging System for an Unmanned Aerial Platform – *Application filed, 2026*
- **Publication:** Entertainment Multimedia Compression Using YOLOv8 and GAN – *Under review, Springer, 2026*

Leadership

Operations Lead

Mar 2024 – Mar 2025

The Bengali Club, VIT Bhopal

- Managed end-to-end logistics and workflows to ensure seamless execution of club cultural events.

Core Member, Marketing

May 2025 – May 2026

Anterix Club, VIT Bhopal

- Executed targeted marketing campaigns and campus outreach to boost club visibility and event registrations.

Certifications & Achievements

- **2nd Place Winner** – Mahakumbh MP 2028 Innovation Hackathon, *Jun 2026* (Kumbh-Cortex Omni-Flow)
- Complete Data Science, Machine Learning, DL, NLP Bootcamp – *Udemy, 2025*